

MINGXUAN (ADRIAN) CAO

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EDUCATION

University of Chicago

Sept. 2024 - Now

Ph.D. in Data Science, Data Science Institute

Research Interest: Biological Generative Models; Reinforcement Learning; High-dimensional Statistics

Washington University in St. Louis

Sept. 2020 - May 2024

AB+AM in Statistics (Honors)

Second Major in Economics and Strategy

Minor in History

RESEARCH INTEREST

My research develops post-training and inference-time methods at the interface of **statistical learning**, **diffusion/SDE generative models**, and **reinforcement learning**—spanning **RL/IL-based fine-tuning** as well as **training-free guidance and alignment**. I aim to enable purposeful scientific discovery in biology and chemistry via scalable algorithms for **planning, sampling, and multi-objective optimization** in complex sequence–structure domains.

PAPERS & REPORTS

Xuefeng Liu*, **M. Cao***, Qinan Huang, Thomas Brettin, Rick Stevens, Le Cong. "Active-GRPO: Adaptive Imitation and Self-Improving Reasoning for Molecular Optimization" [arXiv] **Submitted to NeurIPS 2026.**

Xuefeng Liu*, **M. Cao***, Songhao Jiang, Xiao Luo, Xiaotian Duan, Mengdi Wang, Tobin R. Sosnick, Jinbo Xu, Rick L. Stevens. "Monte Carlo Tree Diffusion with Multiple Experts for Protein Design." [arXiv] **Poster, NeurIPS 2025 Workshop on AI4D3; Submitted to MLCB 2026.**

Qinan Huang, Weizhi Xue, Weizhou Wang, **M. Cao**, Ian Su. "MixUni: Joint Multi-Property Prediction for Chemical Mixtures with Physics-Informed Heads" **Poster, GFM @ ICML 2026; Submitted to NeurIPS 2026.**

Honglin Bao*, Jiawei Zhang*, **M. Cao***, James A. Evans. "From Division to Unity: A Large-Scale Study on the Emergence of Computational Social Science, 1990-2021." **Proceedings of the ACM on Web Conference 2025 (WWW '25).** [arXiv] [Visualization]

Xuefeng Liu*, **M. Cao***, Xiao Luo, Songhao Jiang, Tobin R. Sosnick, Jinbo Xu, Rick Stevens. "Leveraging generative hallucination and biophysics-informed modeling for unified biomolecular sequence–structure co-design" **Submitted to NeurIPS 2026.**

Haojing Zhang, Qinan Huang, **M. Cao**, Rick L. Stevens. "Latent Spectral Dynamics: Uncovering Truthfulness in Large Language Models via Koopman Operator Theory." **In Preparation.**

M. Cao, Xuefeng Liu, Songhao Jiang, Qinan Huang, Ian Foster, Rick L. Stevens. "RL Reasoning with Molecular Design: An LLM-Agent Framework Orchestrating Specialized Foundation Models." **In Preparation.**

M. Cao, Chetkar Jha. "Estimating number of factors in factor analysis for datasets with missing entries at block level." **In Preparation.**

M. Cao, Siyang Wu, Keyao Wu, Honglin Bao, Jiawei Zhang, Aaron Schein, Nikos Ignatiadis. "From Stories to Signals: Multiple-Testing Guarantees for Interpretability Across Foundational Language Models." **In Preparation.**

* Equal contribution.

THESIS

Bootstrap Estimation of Pre-averaged Volatility under Microstructure Noise. *Advised by Prof. José Figueroa-López.*

RESEARCH EXPERIENCE

Agentic AI for Biological Design and Inference

Advisor: Rick L. Stevens

June 2025 – Present

University of Chicago / Argonne

- Developing agentic and inference-time control methods for biomolecular design, including hierarchical multi-agent workflows for IDP-targeting binder discovery under multi-objective constraints such as binding affinity, PPI proximity, scaffold developability, epitope coverage, and structural diversity.
- Building blackboard-style coordination and Pareto-frontier analysis pipelines that integrate specialized design tools, candidate developability prediction, epitope/scaffold diversity tracking, and large-scale binder evaluation on HPC/exascale computing resources.

Interpreting GWAS Signals with Protein/Regulatory Foundation Models

Advisor: Hae Kyung Im and Dan L. Nicolae

Dec. 2024 - May 2025

University of Chicago

- Developed interpretable pipelines to link GWAS loci to molecular mechanisms by combining protein language models and regulatory predictors: (i) used PLM embeddings/channels to quantify how coding SNPs perturb residue environments and functional motifs as pathogenicity surrogates; (ii) modeled noncoding variants with eQTL-informed effect directions and sequence-to-expression predictors (PrediXcan/TFXcan-style, Enformer-like scores) to connect allelic dosage to gene expression and downstream traits while explicitly acknowledging tissue/context specificity and power gaps underlying partial GWAS–eQTL overlap; and (iii) treated selection metrics (e.g., iHS, F_{ST}) as exploratory annotations only, with causal claims relying on LD-aware fine-mapping/colocalization—complemented by a preliminary theoretical result toward fine-mapping without external LD reference panels.

Deterministic Parallel Analysis under Missing at Block Level

Advisor: Chetkar Jha

May 2023 – Present

Washington University

- Extended parallel-analysis factor selection to MAR settings and extending to MNAR by incorporating a shifted Marčenko–Pastur law for eigenvalue thresholds; provided corrections for rank recovery in high dimensions with missingness, and empirically compared against likelihood and spectral baselines across SNR/missingness regimes, showing improved recovery at low ranks.

HONORS & AWARDS

- AIMEMS NRT Team Funding Award, “GeoSolv: Geometric Deep Learning for the Discovery of Non-Flammable Na-ion Battery Electrolytes” (\$10,000) 2026
- Magna Cum Laude, Washington University 2023
- Highest Distinction Graduate in Statistics 2023
- 2023 MCM/ICM Honorable Mention 2023
- 2023 Freiwald Scholars Fellowship in the Department of Math and Statistics of Washington University 2023

PROFESSIONAL EXPERIENCE

Morningstar, Inc.

Investor Relations Intern

June 2023 – Aug. 2023

Chicago, IL

- Analyzed financial and competitive data for quarterly earnings materials, investor communications, and board-level reporting; used data visualization and AI-assisted analysis to extract market and company-level insights.

Shanghai Seeking Sense Investment Management Co., Ltd

Quantitative Investment Analyst

July 2022 – Aug. 2022

Shanghai, China

- Developed quantitative trading and industry-rotation research prototypes using PSY unit-root tests, Bayesian online changepoint detection, and convertible-bond signals; analyzed semiconductor supply-chain valuation and market dynamics.

TEACHING & MENTORING

Teaching Assistant at University of Chicago

- DATA 32100: Generative Models: Theory and Applications (TA) Spring 2026
- DATA 30100: Intro to Data Science (TA) Autumn 2025
- DATA 21300: Models in Data Science (TA) Spring 2025

Teaching Assistant at Washington University

- Math 421I: Statistics for Data Science II (AI & Grader) Spring 2024
- Math 461/5155: Time Series Analysis (AI & Grader) Fall 2023
- Math 459: Bayesian Statistics (UTA & Grader) Spring 2023
- Math 495: Stochastic Processes (UTA & Grader) Spring 2023
- Math 456: Topics in Financial Mathematics (UTA & Grader) Fall 2022
- Math 421I: Statistics for Data Science II (UTA & Grader) Spring 2022

ACADEMIC SERVICE

- Reviewer, COLM 2026
- Reviewer, NeurIPS 2025 Workshop on *AI Virtual Cells and Instruments: A New Era in Drug Discovery and Development*

TECHNICAL STRENGTHS

Computer/Systems	Python , C++ (intermediate), Bash, Linux, Git, CUDA/OpenMP (basic), SLURM/HPC
ML/RL Systems	PyTorch , JAX, TensorFlow, Hugging Face Transformers, vLLM, Ray, Hydra/W&B
Sci/ML Tools	NumPy/Pandas, scikit-learn, RDKit; experiment tracking, reproducible configs, metric guards
Bio/Platforms	ESMFold, AlphaFold, DPLM-2 (ProteinA) , FoldFlow, RFdiffusion ; baseline docking/QSAR
Languages	English (bilingual), Mandarin Chinese (native), French (elementary)